

Technology and Accountability:

Using Blockchain for AI Provenance

In a world of opaque AI systems, provenance; the complete, verifiable record of a model's origins and processing history is the missing link between powerful models and meaningful accountability. Without it, assigning legal responsibility for model outputs is nearly impossible.

The Provenance Gap

Modern AI development suffers from a structural gap where there is no authoritative, tamper-evident record of training data, authorization, or model changes. Current disclosure formats, such as model cards, are largely voluntary, unaudited, and often outdated.

State of AI Training Transparency (2024 Estimates):

- **Dataset name disclosed:** 62%
- **Dataset version/date:** 31%
- **Licensing terms stated:** 24%
- **Filtering criteria documented:** 18%
- **Independent audit completed:** <5%

Dataset Hashing Pipeline: Tamper-Evident Chain

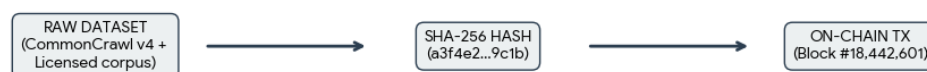


Figure -Error! No text of specified style in document.-1The State of AI Training Transparency

Real-World Consequences

The absence of an auditable trail has led to several high-profile failures and legal disputes between 2021 and 2024:

- **Legal Disputes:** Companies like OpenAI, Stability AI, and Midjourney have faced litigation regarding whether specific copyrighted works were included in training sets that no longer exist in an auditable form.
- **Clinical Bias:** A widely used clinical decision support model influenced treatment for tens of thousands of patients before it was discovered to be trained on demographically skewed datasets with violated licenses.
- **Dermatology Screening:** A model deployed in 2024 performed significantly worse on darker skin tones because its training composition was never publicly disclosed.

Technical Solution: Blockchain Mechanisms

Blockchain offers three specific mechanisms to create a tamper-evident audit trail for machine learning infrastructure:

1. Cryptographic Hashing

Dataset snapshots and model checkpoints are hashed on-chain. Because a cryptographic hash is a unique digital fingerprint, altering even a single byte of data would invalidate all subsequent hashes, proving the state of the model at each training stage.

2. Smart Contract Licensing

Licensing terms for training data are encoded programmatically. These self-executing contracts check compliance and can automatically route royalty payments to data contributors when a model is deployed.

3. On-Chain Registries

Instead of storing the model itself, these registries store provenance metadata: the base model, fine-tuning runs, and dataset versions. This allows regulators to verify lineage (such as GDPR compliance) without exposing proprietary model weights.

The Path Forward: A Maturity Model

As regulations like the EU AI Act and the US Executive Order on AI take effect, the industry must move toward standardized auditability.

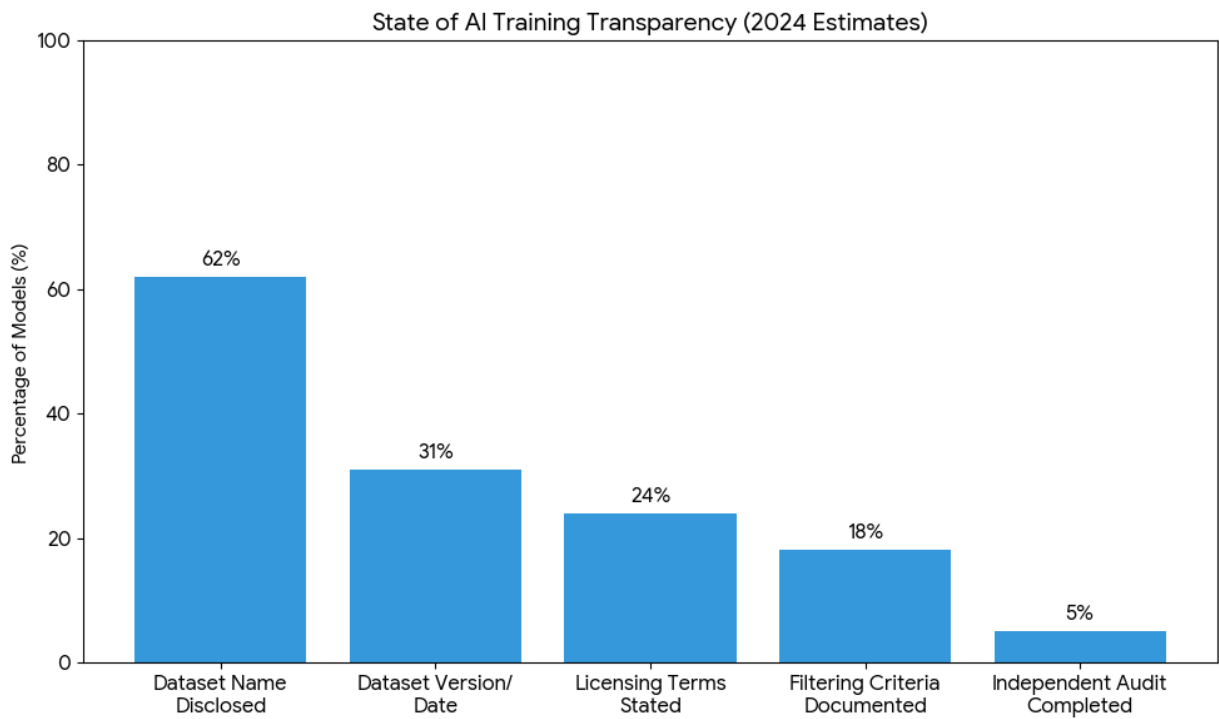


Figure 1-2AI Governance Maturity Model

Stage	Name	Description
Stage 1	Current	Voluntary disclosure via model cards; no verification or enforcement.

Stage	Name	Description
Stage 2	Near-Term	Cryptographic attestation. Dataset hashes recorded on-chain at training time.
Stage 3	Mid-Term	Smart contract enforcement. Programmatic licensing and automatic royalties.
Stage 4	Long-Term	Continuous audit. Deployed models log behavior against registered configurations to detect drift.

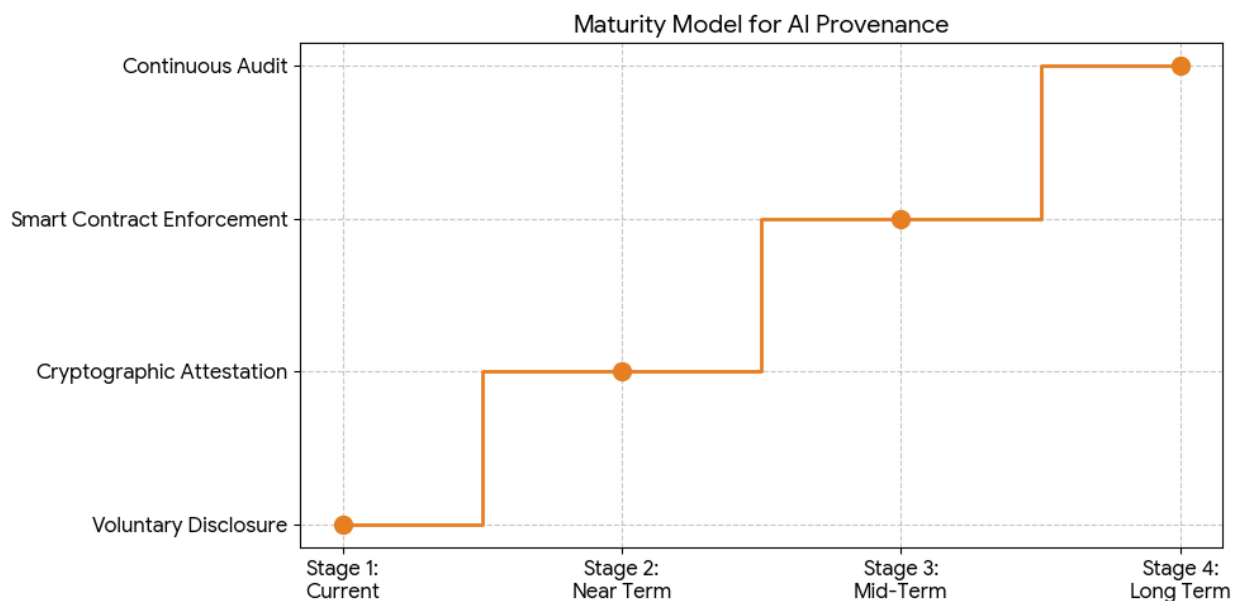


Figure -Error! No text of specified style in document.-3Dataset Hashing Pipeline

Blockchain is not a "magic solution"—it cannot prevent a model from drifting into harmful behavior after deployment. However, it provides the independently verifiable record necessary to move beyond an ecosystem where accountability is determined only by "whoever gets sued last".